HiMCM Contest

The 2025 HiMCM Results

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Congratulations to all teams who participated in the 28th High School Mathematical Contest in Modeling (HiMCM). The HiMCM continues to provide a rich and rewarding experience for students, advisors, schools, and judges around the world. This year, a total of 1,091 teams, each consisting of up to four students and representing 498 schools across 24 countries and regions, came together to engage in this remarkable challenge.

The judges were once again deeply impressed by the dedication, creativity, and mathematical insight demonstrated by participants. We recognize the significant investment of time and resources required for those involved and we extend our sincere gratitude to all who contributed to this year's competition.

We applaud every participant for their work in the HiMCM. Among these tremendous efforts, eight teams rose to the highest distinction and were named Outstanding Winners. In addition, with the support of the National Council of Teachers of Mathematics (NCTM), two teams were honored with the NCTM Award.

Outstanding Teams

- 16390 - Borden International Education School, Hubei, China
- 16442 - Shanghai Nord Anglia Chinese International School (Secondary), Shanghai, China (NCTM Winner – Problem B)
- 16486 - BASIS International School Park Lane Harbour, Guangdong, China
- 16603 - Taipei American School, Shilin, Taiwan China (NCTM Winner – Problem A)
- 16738 - Shanghai Institute School, Shanghai, China
- 16739 - Guangdong Country Garden School, China
- 16843 - Nanjing Foreign Language School, Jiangsu, China
- 17401 - Horace Greeley High School, NY, USA

The 2025 Contest

COMAP remains committed to presenting engaging, real-world problems that challenge and inspire students. As in previous years, teams selected from two distinct problems: Problem A, which focused on modeling Emergency Evacuation Sweeps, and Problem B, which explored the Environmental Impact of Sport Mega-Events.

The 2025 contest received 1091 submissions. Of these, 585 addressed Problem A and 506 addressed Problem B. In total, 4023 students participated, representing a wide range of schools and regions including teams from Australia, Brazil, Canada, China, France, Germany, Hong Kong (SAR) China, Japan, Macau (SAR) China, Malaysia, Norway, Peru, Philippines, Portugal, Singapore, South Africa, South Korea, Taiwan China, Thailand, Turkey, United Arab Emirates, United Kingdom, USA and Vietnam. Within the United States, 230 teams represented 25 states.

Overview

For more than four decades of international modeling contests, COMAP has witnessed growing enthusiasm for mathematical modeling. In 2025, HiMCM marked its 28th year, as well as the 25th anniversary of offering two problem choices. From an initial cohort of just 115 students, participation has expanded and now over 55,000 students worldwide have engaged with complex, meaningful problems through HiMCM. The contest welcomes all high school and below students. Success is grounded not in advanced coursework, but in a solid foundation of high school mathematics and curiosity about solving real-world problems.

We also wish to recognize the invaluable role of advisors and teachers. These dedicated educators inspire students to extend beyond the traditional curriculum and embrace new ways of thinking. Their mentorship is essential to the success and continued growth of the HiMCM, and we are deeply grateful for their commitment.

Rules and Expectations

The core principle of the HiMCM is that all work must be completed solely by team members using inanimate sources such as textbooks, research articles, and other published materials. Collaboration outside the team, including consultation with teachers, experts, or external individuals, is not permitted. Similarly, conducting surveys, interviews,or accessing solutions shared electronically by others violates contest rules. Again, only the team members may contribute to the solution through their knowledge and work. Additionally, COMAP is committed to ensuring that all necessary resources for success are accessible without requiring additional purchases. The materials and information provided

Table 1: 2025 HiMCM Results

Problem	Outstanding	%	Finalist	%	Meritorious	%	Honorable Mention	%	Successful Participant	%	Unsuccessful	Disqualified	Not Judged	Total
A	5	1%	21	4%	80	14%	158	27%	315	54%	1	2	3	**585**
B	3	1%	27	6%	63	12%	152	30%	255	50%	0	5	1	**506**
Total	8	1%	48	4%	143	13%	310	29%	570	52%	1	7	4	**1091**

by COMAP, along with a team's knowledge, skills, and research, are all that is needed for success.

The use of artificial intelligence (AI) is permitted in limited, transparent ways. For example, AI may be used for research assistance, editing, or translation. All AI use must be fully documented, including prompts and outputs, in a report that does not count toward the page limit.

Advisors and students often ask what level of mathematics is required and what special coding skills are needed for the contest. Our HiMCM problems are accessible using high school level mathematics alone and no programming or coding skills are required. As in all our contests, each of our problems is accessible on multiple levels, with many teams making use of different levels of mathematics and technology tools. Students should apply the mathematics they understand and are able to explain in their solution analysis, as the ability to clearly communicate their methods and rationale is valued in final judging.

COMAP understands that there is not one standard or body of knowledge of high school level mathematics for an international math modeling competition. The judges see a wide range of abilities and skills. Teams should note that using lower levels of mathematics properly can do just as well as higher levels of mathematics in the competition. Teams should not be focused on impressing the judges with overly complex techniques as models but rather concentrate on using good modeling techniques in putting together their solution.

COMAP uses X and Weibo to provide contest information to participants. Follow us @COMAPMath on X or COMAPCHINAOFFICIAL on Weibo for contest guidance and up to date contest information.

Judging Process

Judging is conducted blindly without identifying information about teams or schools. As open-ended problems, submissions are evaluated on their assumptions, reasoning, and modeling approach rather than adherence to a single correct answer with a rubric. All submissions are reviewed electronically, allowing for a diverse and highly qualified judging panel to easily review. Each paper is initially evaluated by at least two preliminary judges. We extend our sincere thanks to these judges for their thoughtful and thorough assessments.

After the preliminary round, papers are categorized as Finalist, Meritorious, Honorable Mention, or Successful Participant. Finalist papers advance to a final round of judging, where the most exceptional submissions are selected as Outstanding. This year, 105 papers advanced to final judging, with eight ultimately earning the Outstanding designation.

Looking Ahead

The HiMCM continues to evolve as a platform for students to apply mathematics in meaningful and collaborative ways. Additionally, COMAP has expanded opportunities through the MidMCM, now in its fifth year, offering middle school/level age students the chance to engage in mathematical modeling. Both contests emphasize the power of mathematics as a tool for understanding and representing the world. Students develop critical thinking, communication, and teamwork skills while gaining confidence in tackling complex, open-ended challenges. Importantly, no advanced mathematics or programming experience is required. Success lies in clear reasoning, effective communication, and thoughtful application of accessible mathematical tools. Judges value clarity and sound methodology over unnecessary complexity.

We encourage advisors to continue fostering creativity, curiosity, and perseverance in their students. COMAP encourages all mathematics faculty to get involved, motivate your students to be problem solvers, make mathematics relevant, and open the doors to future success. We want to partner with teachers as we continually strive to improve the contest and make it accessible and impactful to all students. Together, we can ensure that mathematical modeling remains an engaging, accessible, and impactful experience for all.

2026 Contest Dates

Mark your calendars for the 6th annual MidMCM, and the 29th HiMCM, to be held November 4 - 17, 2026. Registration for the 2026 MidMCM and HiMCM will open in September. Student teams may work at any time during the contest window. At the team members' convenience, teams download and choose their problem, complete their modeling solution, and electronically submit their solution document by the deadline on November 17th. Again in 2026, one team for each problem will receive the NCTM award. Teams can learn more about COMAP's contests and registration at www.comap.org/.

MathModels.org Has Moved

Powered by COMAP content, MathModels.org is a wonderful resource for students and teachers to make mathematical modeling a year-round activity. This site includes past competition problems, exemplar solution papers, and judges' commentaries that can provide guidance for teams preparing to compete. Teachers and students may use the materials found on this site to enrich their classes and help prepare students for mathematical and interdisciplinary modeling contests. We encourage you to visit this wonderful site in its new home on the COMAP webpage: www.comap.org/mathmodels.

MATHmOdels.org

The following guidance reflects the qualities our judges consistently value in HiMCM papers. Strong submissions present a clear and comprehensive summary and address all aspects of the problem through the thoughtful development and application of one or more mathematical models. The most competitive papers distinguish themselves through clarity, organization, and well-supported reasoning, demonstrating both careful analysis and effective communication.

The strongest papers go beyond meeting the basic requirements. They present complete and well-integrated mathematical and logical analysis in a cohesive and engaging manner. These submissions are not only thorough but also accessible. They are easy to follow, logically structured, and often creative in their approach. They include clearly stated assumptions with appropriate justifications, a transparent description of the modeling process, well-explained results and analysis, and thoughtful discussion of strengths and limitations, including sensitivity analysis. Strong conclusions and appropriate references further support the overall quality of the work.

Problem Discussions

Our judges continue to emphasize that HiMCM problems are designed to be accessible using high school-level mathematics. While some teams attempt to incorporate advanced concepts or tools, these approaches are only effective when they are clearly explained and appropriately applied. Papers that rely on techniques beyond a team's ability to fully justify or interpret are often less successful. We are not seeking the use of the most advanced mathematics. Rather, the most successful papers are those in which teams develop models grounded in mathematical ideas they understand, can clearly explain, and can apply effectively.

The specific problem discussions below provide comments on the problem and how teams addressed the requirements. In the next section, we provide judges' comments about the solutions by breaking down the various parts of a submission and providing exemplars. To view the complete problem statements, visit www.comap.org/mathmodels or www.himcmcontest.org.

Problem A: Emergency Evacuation Sweeps



For the first time 25 years ago, COMAP posed two problems for High School students to choose from during the HIMCM. To honor this anniversary, we opted to revisit Problem A but with a different scenario. In 2000, student teams were asked to develop an efficient algorithm for police cars to sweep the area to force bank robbers (who were fleeing by car) into one of the established roadblocks. Today, we challenge teams to conduct a different type of sweep.

During emergencies such as fires or gas leaks, rapid and orderly evacuation of a building is critical for saving lives. Trained personnel responding to these emergencies (responders) often sweep a building – going from area to area and room to room – to ensure all occupants have evacuated safely or are actively moving towards the exits. This would include visually assessing, if possible, all spaces to confirm that they are empty with the goal of guaranteeing no one is left behind. The efficiency and strategy behind these sweeps could mean the difference between safety and true peril.

Teams were asked by the Clearing Of Multi-use Assembly Public/Private-spaces (COMAP) with designing a mathematical model to optimize sweeping strategies in multi-floor buildings during emergency evacuations. The goal is to ensure all rooms are cleared in the shortest possible time, while prioritizing occupant safety and responder efficiency. Buildings vary in layout, number of floors, stairwell placement, location of exits, and occupancy patterns while responders face pressure to act quickly with sometimes limited information. These considerations must be included in your analysis.

This problem makes teams look at emergency evacuation from different perspectives. Students have been taking part in fire drills in school throughout their lives; now they can see it from the other perspective while considering both the

buildings themselves and the responders in all of their modeling decisions, applied to three different layouts/scenarios.

Teams were asked to build a sweep model after conducting research on the aspects that they would need to incorporate. The sweep models were then applied to a simple building layout before extending the model with additional complications and applying it to different types of buildings. Teams were then asked to draft a formal letter to a local Emergency Planning Committee addressing both how to sweep buildings and assign responders including recommendations for strategies that the committee could use to classify different types of building and how to respond.

Problem B: Environmental Impact of Sport Mega-Events

Every year, the Super Bowl brings together millions of people – from fans traveling to the host city, to viewers tuning in from around the globe. This event generates enormous social and economic value, but it also comes with a significant environmental cost. With air travel, hotel stays, energy use, food consumption, waste generation, and massive broadcast operations, the Super Bowl's environmental footprint is substantial.

As we grow more concerned about sustainability, large-scale sporting events or sport mega-events like the Super Bowl are being made more environmentally responsible. External organizations have developed models and scoring systems to rate the sustainability of these large-scale events based on many factors from energy use and waste management to transportation and carbon offsets. While these environmental scores exist and a wealth of data is available on what event planners are doing to be eco-friendly, sustainability rarely drives key decisions such as where to host an event or how to design its required supporting infrastructure. This disconnect presents an opportunity to build a model where ecological impact is not just an afterthought, but a core driver of planning and location decisions. Imagine a scenario in which host cities are evaluated not only for their economic viability and fan capacity, but also for their renewable energy potential, existing infrastructure efficiency, and transportation accessibility. Such a model would enable organizing committees to balance performance, cost, and sustainability in a measurable way.

The Consulting Organization Making Athletics Planet positive (COMAP) challenged teams as sustainability consultants to assess and reduce the environmental footprint of sport mega-events. The goal is to understand the ecological impact of these events and explore how changes in event planning could make a difference. This problem is not simply about creating a model to assess the environmental impact of large-scale sporting events or sport mega-events. Instead, the focus was on bringing environmental impact to the forefront of the decision-making on where to host.

Teams were asked to build a decision model after conducting research on factors included in current Super Bowl sustainability Metrics. The decision model was to be applied to where to host a future Super Bowl before extending the model with additional complications and applying it to a different type of sport mega-event. The addition of infrastructure in this extension was a key factor. Teams were then asked to draft a persuasion paper letter to the NFL recommending the city that should host the Super Bowl because of sustainability.



Judges' Discussion

Here we will examine in-depth what judges were looking for in each aspect of a well-written submission. For each piece, we will outline what we expected to see, possibly including some specifics related to each problem, and then highlight actual examples from this year's Finalist and Outstanding papers. Students and coaches are encouraged to review these discussions and incorporate these best practices in future competitions.



Note!
An extensive online Appendix "2025_HiMCM_Examples" containing all the examples mentioned below can be found here:
www.comap.org/2025HimcmExamples

Overall

Student participants are expected to ensure that their papers adhere to all contest rules as outlined on the official contest website. Submissions must be presented within the 25-page limit and use a minimum 12-point font. Each year, judges encounter papers that attempt to exceed these limits by reducing font size, adjusting margins, or adding additional pages. While such submissions are not automatically disqualified, they are unlikely to earn top designations.

Papers that advance to the Finalist and Outstanding levels create a positive and lasting impression and consistently demonstrate a high degree of clarity, organization and coherence. They begin with a clear and informative summary and provide a structured preview of the paper. These submissions present well-justified assumptions, clearly explain the development and application of their model, and support their results with appropriate mathematical analysis. The strongest papers also include sensitivity analysis, discuss model strengths and limitations, and conclude with clear, well-supported findings. Ultimately, it is not the length of a paper but the clarity, completeness, and effectiveness of its communication that distinguishes strong submissions.

Before submission, teams are encouraged to carefully review their work, including the use of spelling and grammar tools. For international teams, all symbols and labels in tables and figures should be presented in English. To ensure fairness in the judging process, student and school names must not appear anywhere in the paper.

Executive Summary

The executive summary is an essential component of a standout modeling paper as it requires synthesizing the most critical information from the report into a clear, concise overview of the work to follow. A memorable executive summary sets the tone for what can be expected throughout the report, while also being able to stand alone as a complete depiction of the problem from its introduction to its conclusion including implementable recommendations.

Teams are encouraged to write the Executive Summary after completing their paper so they can accurately capture the entire analysis. Once written, it should be reviewed carefully to ensure it stands alone as a representation of the full submission.

For Problem A, Team 16486's executive summary (Example 1) found an effective mix between providing high-level, lay-person overview of their model while including technical keywords to prime the reader as to what the implementation would entail. They include numbers quantifying the performance of their model that are understood in the context of the problem statement and finish their summary with a concise description of their key contributions.

For problem B, this meant addressing both the Super Bowl and another global mega-event and addressing the key environmental impact factors chosen to develop the model around. An excellent example of this is from Team #17173 (Example 2). Despite being one of the shortest executive summaries submitted, it efficiently and clearly summarized the goals, methods and results of their work in a way that was distinct from the main body of the report. This summary whets the appetite of the reader, enticing them to read further to learn more about how the locations for the events were chosen.

Demonstrated Contextual Understanding

Strong papers demonstrate a clear understanding of the complexity of the problem beyond what is explicitly stated in the prompt. This understanding is often reflected in the introduction, where the better teams move beyond simply restating the problem to provide their own unique meaningful insight into the bigger picture. Some teams incorporate background research or elements of a literature-style review, though this is not required. Teams should know that what matters most is the depth of understanding conveyed. This deeper perspective strengthens the entire paper. It enables teams to more clearly define the problem and make thoughtful modeling decisions.

Teams are encouraged to make the introduction their own by establishing a strong foundation before beginning their model development. Judges

appreciate well-developed insights, even those that are not directly incorporated into the final model, as they demonstrate deeper engagement with the problem. Clear documentation of research and reasoning further enhances the credibility and quality of the work.

This year and in the past, judges looked for teams that provided well thought out insights into the problem. Sometimes these insights had to be ignored to build the base model, but the fact that a team would recognize and communicate these insights was appreciated.

Additionally, some teams produced a literature review. Using outside references to provide understanding of the problem was resourceful. Some teams did great without the literature review. Judges looked for teams that produced insights and understanding of the problem that went beyond what was in the problem statement.

Judges saw many excellent examples of what demonstrated contextual understanding can look like in a submission. Team 16969, when tackling Problem A, produced a well referenced literature review (Example 3) that discussed the multitude of problems and considerations that should be examined when tackling an evacuation problem. Additionally, judges read many papers that provided a good understanding of the problem when they went above and beyond regurgitating the problem statements in their introduction. As an example, Team 16603 (Example 4) discusses the condition for an ``all clear'' to be given in an emergency evacuation. Papers such as this tell the narrative of what the objectives are for addressing the situation. In Problem B, Team 16521 (Example 5) conducted a literature review and then nicely consolidated this research in their section entitled "Problem Analysis". Here the team explained how the reviewed foundational works on the environmental assessment of sport mega-events and how that research drove the key environmental impact areas that they chose to focus on in their modeling. This strong foundation directly set the stage for their model development.

Judges would also like to highlight a team that demonstrated that they truly understood why they were tackling this problem in a different way than most. Team 16739 (Example 6) had a section towards the end of their paper, after they had implemented the model for the Super Bowl, adapted it for multi-sport or multi-venue events, and then implemented that model for Summer Olympics. The team discussed key differentiating mechanisms between types of sport mega-events in a section entitled "Comparative Analysis of Model Results" which truly demonstrated to judges that they understood why they were conducting the analysis they had. This section was not a problem requirement but instead demonstrated this team's deeper understanding of the bigger picture problem which was to determine what factors matter the most for different types of sport mega-events when it comes to determining the most sustainable host cities.

Assumptions

Teams are expected to clearly state all assumptions, either as a dedicated list or within the context of the paper where they are applied. Assumptions play a critical role in simplifying complex real-world problems and addressing gaps in data or knowledge. Effective assumptions are purposeful and directly connected to the model. Each assumption should be clearly justified, with reasoning that explains both why it is necessary and how it supports the modeling approach. Judges look for assumptions that are thoughtfully chosen and meaningfully integrated into the analysis, rather than included as general statements.

Teams are encouraged to consider where and how each assumption is used within their model. The strongest papers not only state and justify their assumptions but also demonstrate explicitly how those assumptions influence the development, application, and interpretation of the model.

Judges for Problem A looked for teams that used assumptions as a tool to assist in building the base model for searches to eliminate complexities so the base model could be built. Those teams would then build further models without the assumption constraints of the base model. For example, in the paper written by Team 16486 (Example 7), the team clearly states an assumption that will be in place for the base model. The team later builds upon the base model when assumption 5 is no longer in place.

Additionally, judges looked for papers that referenced where they obtained their data for certain assumptions. The judges appreciated when teams clearly referenced where they obtained the speed they used for the movement of firefighters like in Team 16691's paper (Example 8).

In Problem B, the paper written by Team 17205 (Example 9) was very readable, but they did an exceptional job of making their assumptions clear by organizing them into categories.

Model Selection

There are many valid approaches to modeling HiMCM problems, and judges appreciate appropriate and creative solutions. The most successful teams identify the key factors driving their model and explain the reasoning behind their chosen approach. This includes discussion of model inputs, structure, and data, as well as any adaptations made when using existing models. Strong papers often acknowledge the distinction between an ideal model and what is feasible given available data and time constraints.

While teams may explore multiple approaches, successful papers present a clear and focused modeling strategy. Rather than introducing several disconnected models, strong submissions develop a primary model that is refined or extended as needed. Judges consistently favor well-reasoned, clearly applied models over overly complex approaches that are difficult to justify or implement.

While the problem statement for Problem A did not make direct mention of graphs or networks, almost every team used a network model as the underlying framework. How these

graphs were adapted to finding optimal paths varied among the submissions though, with some papers using a linear programming formulation, some papers using optimal graph path algorithms like A*, traveling salesman, or Dijkstra, and/or some papers incorporating stochastic simulations to explore these relatively large state spaces.

One common approach was to formulate room allocation and path planning in one swoop using a linear program or similar optimization framework. Team 16486 (Example 10) used this approach, effectively describing their set-up in a direct manner. They formulated the sweep-plan as a room and edge assignment problem and described several constraints their solution should meet, which encoded that a sweep-plan must travel along consecutive edges and every room must be swept, among others. With these determined, they then naturally define their objective function, which allocates all room assignments and responder paths in one swoop. In their model evaluation, they also considered three other strategies for room assignment, showing that some thought went into deciding to use a linear program.

In slight contrast, many teams opted to build and describe their models in a way that highlighted properties of the networks a little more explicitly. This is not to say one approach is better than another, but rather the tools that become applicable are more easily seen in one versus another. This is the approach Team 17401 (Example 11) used, splitting the room allocation and path-finding problems into two pieces, and using graph algorithms for the second piece. In their model, rooms of a building were first clustered based on expected workload to clear them, after which an optimal path was computed using a traveling salesman problem solver they built and adapted to this problem. Team 16603 (Example 12) used a similar approach as above, splitting the problem into a room allocation and then path-finding problem, though in this case they used existing traveling salesman solvers for the initial path construction. The networks they constructed were dynamic, so it could happen that a responder's original path is no longer available; in these instances, their model used an A* algorithm to find a way around these obstacles. Moreover, they effectively used Monte Carlo simulations for the dynamic graphs to gather data and provide strong evidence for the optimality of sweep-plans they construct in the presence of noise.

These last two models naturally followed from their approach to the problem: first identify who should search which rooms, then figure out how to get them there. This contrasts with the first model mentioned, in which the two steps are performed at the same time. As stated earlier, neither approach is better than another - teams should explore the problem and potential solutions and let their work tell them which approach makes the most sense.

For Problem B, there were many modeling options and techniques available to teams participating in this competition, although many teams utilized similar approaches. What set teams apart was their ability to not only describe what they did, but why they used the approaches they chose. For example, many groups this year used Analytical Hierarchy Process (AHP) and Entropy Weight Method (EWM) together for a multi-criteria decision-making model and justified balancing the subjectivity of one with the inherent objectivity of the other. The main factor distinguishing groups using similar approaches was the ability to justify how the rankings were determined, or how the balance between methods was determined for their multi-criteria decision-making model. There were some excellent papers submitted that did not use the most powerful or unique modeling techniques but distinguished themselves by clearly and effectively explaining the reasoning behind their choices. For examples, judges chose to present the three Outstanding papers as they all had more standard modeling approaches with excellent extensions and explanations.

For their multi-criteria decision-making model, the Outstanding Team 16442 used AHP and EWM to determine the weights of their indices. This team used five core indices of environmental impact based on over 14 contributing factors. This allowed them to create their City Impact Evaluation Model (CIEM) to evaluate potential host cities for the Super Bowl followed using Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) for the determination of a sustainability score which can then be ranked. What judges appreciated the most about this standard approach was the team's explanations of the process and their adaptation of this method for their model extension (Example 13). When Team 16442 went to determine the best potential host countries for the World Cup, they adapted their initial CIEM by refining the five core indices and then added one more index to cover the environmental impact of infrastructure.

The Outstanding Team 16738 chose a similar methodology for their multi-criteria decision-making model. They utilized 12 secondary indicators and then Analytic Network Process (ANP) with EWM followed by TOPSIS. The team explained their choice of ANP over AHP (Example 14) because they wanted to capture the interdependencies between criteria. The judges appreciated the analysis that followed which allowed the team to extrapolate into the future based on what they determined would be the two indices that would most likely change significantly over a five-year period due to newly established public policy (Example 15). It was insightful for the team to note and understand that a sustainability score for 2029 might be significantly different than today's score if certain city's trends in implementing more green practices were considered.

Finally, the Outstanding Team 16739 approached their model creation a bit differently, still using techniques that are regularly seen by judges throughout this and other contests. The team chose Principal Component Analysis (PCA) but then took it further based on Pearson correlations and then a PCA-based multi-variate linear regression model (Example 16) to identify grid carbon intensity, renewable energy share, public transit coverage and the share of green or low-carbon travel as the main drivers. The next step for Team 16739 is what the judges really appreciated as the team used Autoregressive Integrated Moving Average (ARIMA) models to forecast future city characteristics for both cities that have posted and three that have not yet hosted. The identified features were then inputted back into the original analysis before conducting EWM.

Model Implementation

Teams must apply their model to generate meaningful results and demonstrate how those results address the goals of the problem. This requires more than presenting outputs as teams should interpret their findings in the context of the model's purpose and explain how the results inform the original questions. Strong papers reflect thoughtfully on their results, considering whether the model effectively captures the key aspects of the problem and whether the outcomes are reasonable and actionable. Once the theoretical development of their model is developed, teams are expected to apply their work to the problem to get tangible results that address the problem statement. In Problem A's case, teams were expected to implement their models to analyze response times for a simple six room office building and then consider at least two more building layouts.

Many teams defined a dynamic weighted graph representing the geometry and logistics for sweeping a building, allowing for dynamic hazards and the subsequent effects on the network and first responders' possible sweep-plans. Team 16843's implementation (Example 17) stood out to the judges for the in-depth and explicit implementation framework for their optimal paths, which took shape as five algorithms written in pseudocode accompanying their discussion. This helped bridge the gap between theory and implementation, which made analysis of their model's results easier to digest and understand.

Additionally, Problem A judges enjoyed the creativity displayed by teams in the variety of examples considered with their two additional buildings, especially when teams used building layouts based on buildings relevant to their lives. This was one of the reasons Team 17402 stood out during final judging: they considered actual layouts for a nearby kindergarten school (Example 18), as well as a floor of a high school building (Example 19), and performed an extensive analysis of their model applied to each (resulting in an optimal number of 3 and 4 firefighters to sweep the kindergarten and high school floor respectively).

In Problem B, the first application was to finding the most sustainable host city for the Super Bowl. Many teams simply took their model and ran it to determine the best host city for 2029. Judges wanted to highlight two teams that went further on their implementation by providing a differentiation and expansion that made their analysis more trustworthy and demonstrated to judges that they understood the details of this bigger problem. Team 16947 (Example 20) understood the data differences between teams that have previously hosted the Super Bowl and teams that have never hosted. This creates difficulties in the model when attempting to determine who should host in the future. This team implemented them separately and then nicely brought the lists together for their results discussion. Judges also appreciate the work that Team 17173 (Example 21) conducted to add time adjustments. They recognized that cities hosting in earlier years might obtain lower scores using their model because of technological limitations. As they did not want the temporal effect to penalize host cities from the earlier Super Bowl years, they implemented this time adjustment methods after the initial model results and then compared the differences. They then appropriately made their determination based on the updated model.

Model Cohesion & Integration

Teams should guide the reader logically from assumptions to model development, application, and results, ensuring that each step is connected. Papers that demonstrate this coherence are the most effective. While teams may analyze multiple aspects of a problem, the greatest value comes from integrating these components into a cohesive narrative that addresses the overall question. Strong papers demonstrate how the requirements connect, whether by building upon an initial model or adapting it to address additional dimensions of the problem. When multiple models or approaches are used, teams should clearly justify these choices and explain how each contributes to a deeper understanding of the problem. The most successful papers move beyond isolated analyses to present a connected line of reasoning that adds meaning and insight to the bigger picture problem teams are trying to solve.

In Problem A, teams overwhelmingly adapted graph-based approaches to build their model and apply it to the scenarios given in the problem statement. Judges particularly appreciated the discussion in Team 16390's development (Example 22), wherein they built their model while including pseudocode and enough details that it was clear how the model was to be applied, and how their unique approach to task allocation was to be integrated in their graph search and sweep model. This was followed by a brief model validation, then presentation of results, and finally a discussion of sensitivity. They presented a cohesive model discussion (from development to results) in this section, which gave judges a clear view of their approach for a single level building before extending their model to multiple floors, and other more complex scenarios.

Problem B judges viewed this aspect critical to identifying the best papers. If a team was able to integrate all their modeling into a cohesive narrative where subsequent work is linked and dependent on the initial work, then they were awarded a higher designation. The three Outstanding teams, Team 16442, Team 16738, and Team 16739 were able to do exactly that. Judges would also like to highlight the work of Team 16831 who had unique constructs for analysis of Super Bowl hosts where they examined those that have hosted, will host and have not yet hosted as well as the sustainability factors pre-event, during the event and then post-event. Additionally, the judges truly appreciated the approach Team 16831 (Example 23) had when developing their multi-venue, multi-day model. The team used the same model and added new sustainability factors that mattered with the change in scope like transportation and infrastructure. This idea of taking an initial model and then continually refining it to add the complexities that a team wishes to analyze is highly sought after by judges.

Sensitivity Analysis

Sensitivity analysis is an important component of the modeling process, as it examines how changes in assumptions, parameters, or input data affect the results of a model. It helps determine the robustness, reliability, and overall usefulness of the model. Effective sensitivity analysis often includes revisiting simplifying assumptions and evaluating how variations in key inputs influence outcomes. While there is no single required method, judges look for clear rationale behind the approach taken and thoughtful interpretation of the results.

Strong papers use sensitivity analysis to justify the level of confidence in their model and its conclusions. While some teams present this analysis as a dedicated section, those that integrate it throughout their work, connecting it directly to model development and implementation stand out. Judges feel that a well-executed sensitivity analysis demonstrates a deeper understanding of the model's behavior and its practical implications.

Solutions to this year's Problem A, Emergency Evacuation Sweeps, included a variety of factors and objectives. In reflecting on the stability of their models, most teams needed to consider the relative importance of a number of parameters describing the effects of such things as the walking speed of responders, changes due to the presence of smoke or clutter, or the number and/or condition of possible victims in the building. In general, such explorations involve systematic variation of parameter values so that the influence of that parameter may be isolated. Ideally, such influences would then be compared and interpreted in the context of the overall objective of the emergency sweep model.

Team 17460's sensitivity analysis discussion (Example 24) of its process is particularly strong. They describe exactly how the analysis was conducted, identifying the parameters to be tested, the range of values to be used, the method of testing, and the key output of interest, all clearly enough that the reader could duplicate their work independently. They report their results in easy-to-read, easy-to-compare graphical form, and the final discussion provides practical implications of these comparisons for planners for emergency response teams.

Judges from Problem B were looking for an extensive sensitivity analysis as there were many assumptions teams should make, data that had to be researched and incorporated, as well as a variety of parameters for most models. The presentation of the sensitivity analysis by Team 16442 (Example 25) was notable for its depth followed by the explanations of the meaning. The team conducted two examinations, and each was followed by graphics with results and then an analysis of results. Additionally, judges were impressed that the team brought it all together with a conclusion that reinforced their confidence in their selection for the most environmentally sustainable choice for the Super Bowl host city in 2029.

Strengths, Limitations and Conclusions

After completing the modeling and analysis, teams should include a clear discussion of the strengths and limitations of their model, along with concluding insights. This process pushes students to step back and look at the bigger picture of what their model can achieve and where there may be room for improvements, which is entirely reasonable and expected given the time constraints of the contest. It is easy to fall into the trap of saying "Our model is amazing and can solve any problem!" but judges are looking for a realistic assessment. Effective discussions highlight what the model does well by identifying the types of questions it is best suited to address while also acknowledging its limitations. These limitations may stem from simplifying assumptions, data constraints, or modeling choices. Strong papers use this section to reflect on the entire modeling process, offering insight into what worked well, where improvements could be made, and how the model might be extended with additional time or data. In many cases, judges are already familiar with the strengths and limitations of a particular modeling approach, and it is refreshing to see teams arrive at these same conclusions independently.

Clear, well-supported concluding thoughts that tie back to the problem and results are essential, ensuring the paper ends with a sense of purpose. With so many demands asked for in a modeling problem like this, it is easy to overlook the importance of including a conclusion in the report. The conclusion should contain direct recommendations as we solve these real-world problems to provide insight to interested parties. Team 16691 in Problem A (Example 26) did

an excellent job highlighting the capabilities of their model as it represented real safety and rescue priorities. They also chose robust optimization tools. This team also gave an honest reflection by relating their limitations directly back to specific assumptions. The judges also appreciate that their strengths and weaknesses were not presented as a bullet list but written out clearly as a cohesive discussion. It gave the impression that the students really took time to justify each of their assumptions and saw the connection to their model's value.

In Problem B, Team 16624 (Example 27) recognized that the environmental performance of a city does not remain stagnant from 2023 to 2029, as assumed in their model. The team acknowledged that future environmental performance could be forecast by incorporating data from 2023 and earlier. They identified the inclusion of planned sustainability upgrades for a city as a direction for future work.

Non-Technical Document

The purpose of a letter or other non-technical document is to communicate model-based recommendations in a way that is accessible to a non-technical audience. This section demonstrates a team's ability to translate complex mathematical analysis into practical, understandable insights. Since the intended audience often lacks technical expertise, teams must present their findings in clear, concise language, using an appropriate format such as a letter, memo, infographic, or article, as specified in the prompt. Rather than simply stating general practices, strong submissions explain what the model reveals about the implications of a particular course of action.

While some teams may think of it as a secondary component, the non-technical document can be a distinguishing feature of a strong paper. The most effective submissions reflect a clear understanding of the modeling results and communicate them in a way that is both informative and actionable for decision-makers. Judges thought that Team 16846 effectively conveyed their approach to non-technical audiences for Problem A (Example 28), which was amplified by the team publishing the letter at the start of their paper right after their executive summary. They provided a brief overview of their approach before spending the most space discussing their results in context, including some diagrams of actual plans produced by their model.

For Problem B, judges really focused on the omission of specialized mathematical terminology and the avoidance of acronyms without clear definitions. Because the letter is addressed to executives in the NFL rather than to students or instructors in mathematical modeling, accessibility and simplicity were essential. Many papers met or exceeded this standard of proficiency.

However, Team #16739 (Example 29) provided one of the strongest templates for a non-technical letter. The letter began with a concise summary of the model considerations, followed by the recognition of the winning city. To persuade the NFL of their recommendation, the team explained the reasons for their choice in clear, common language rather than technical detail. The letter concluded by identifying other potential host cities and offering an inspiring rationale for making environmental impact an important factor in the selection of a host city.

Appropriate Explanations

Discussions throughout the paper should be clear and detailed to allow the reader to follow the team's modeling process from beginning to end. This includes not only describing what was done, but also explaining why choices were made, with connections between modeling decisions and the requirements of the problem. Every year, judges seek to highlight submissions that offer a balance of sound mathematics with well written justifications of their approach. Judges looked for implementable measures that were backed by research and then well explained. The Emergency Evacuation Sweeps problem of Problem A is closely related to the well-known Traveling Salesman Problem of graph theory. Essentially all student solutions to Problem A began with converting a floor plan and frequently other important information into a vertex-edge graphical representation. Starting with such a representation the task then becomes two-fold: (i) generate a set of paths that covers all rooms, and (ii) minimize an appropriate "cost" function associated with such paths. While it is possible to formulate a "brute force" solution to these tasks, such an approach becomes computationally impractical very quickly. The examples below illustrate (portions of) two different approaches to addressing this difficulty. In each case the teams made it easy for the reader to follow their work, using clear language to explain and provide essential information such as values of key parameters and references for further information.

Team 17401 (Example 30) addressed the computational difficulty by breaking their task into two steps, first partitioning rooms into localized groups, then building appropriate paths. For each task, the students provided clear instructions on exactly how the work is carried out. In addition to these instructions, they provided citations for each step of the process, making it possible for the reader to carry out their work independently but also to find additional theoretical discussion of the methods used. Team 16691's approach (Example 31) to solving the problem of computational complexity uses Ant Colony Optimization, a probabilistic simulation. Even though this approach may not be as familiar as more direct methods, the students provided first a general overview of the ideas behind the method, then gave the details of their actual set-up, including their selected parameter values and stopping rule. The reader would easily "follow along" with the solution process. Again, as is the case in Team 17401's solution, reference to the method is cited.

For Problem B, judges believe that what set teams apart was their ability to not

only describe what they did in their modeling process, but why they used the approaches they chose. There were some excellent papers submitted that did not use the most powerful or unique modeling techniques but distinguished themselves by clearly and effectively explaining the reasoning behind their choices. In addition to narrowing down the key factors driving their model, Team #16738 (Example 32) did an excellent job of highlighting the disparity between potential host cities' performance on these different benchmarks to emphasize the reason for their inclusion in their model.

Organization & Writing

Strong papers are well-structured, clearly written, and easy to follow, guiding the reader smoothly through the team's approach and analysis. A clear organizational framework allows the reader to understand how each component of the paper contributes to the overall solution. Effective submissions consistently connect assumptions, methods, and results back to the context of the problem. The strongest papers enhance understanding through thoughtful presentation. Equations are paired with clear explanations, and visual elements such as graphs or tables are used when appropriate to communicate complex ideas. Overall, strong communication demonstrates a deliberate effort to make the modeling process accessible to the reader.

Team # 16603 (Example 33) submitted a well-organized and well-written paper for their solution to Problem A. Note that even though it was outlined similar to the template, it had a natural flow between sections and information was presented in a reasonable order. One judge in particular appreciated the way that the definitions of the variables were well-organized with descriptions of why and how they were used with units included. When they presented a relationship between variables or a piece of their model, they explained why it made sense. They also summarized all of their findings together and included sections on validating their model. In Problem B, papers that impressed the judges had clear explanations of the models. Judges valued knowing why choices were made. Explaining abbreviations is helpful to the reader. When diagrams or graphs were used, the explanations of them should immediately precede or follow the graph or diagram. Judges appreciated papers that had been proofread and were free of spelling errors or incomplete sentences. We recognize that teams used AI platforms to help with translation and clarity, but teams should make sure their final papers use authentic language. Team #16442 (Example 34) wrote a very readable and organized paper throughout.

Tables & Figures

Effective visualizations, such as charts, graphs, and diagrams, play an important role in enhancing understanding and supporting key points within a report. When used thoughtfully, they can clarify complex relationships and make results more accessible to the reader.

However, visualizations should not stand alone. Each must be clearly explained within the body of the paper, with explicit discussion of what it shows, why it is included, and how it supports the team's analysis. Tables and figures should complement the written analysis rather than replace it, helping to communicate complex ideas more clearly and reinforcing the team's reasoning and findings.

For Problem A, the paper written by Team 16486 had so many exemplars for figures. The floor plan conversions in Figures 3, 5, and 7 (Example 35) were clear, showing side-by-side comparisons of architectural layouts and their corresponding node-graph representations with explicit node labels that make the abstraction process transparent and reproducible. Figure 6 and Figure 8 (Example 36) which were the optimal sweep routes for the school and house layouts, made effective use of color-coding, assigning distinct colors to each responder team's route, clarifying how the optimizer naturally partitions buildings into intuitive zones. Table 3 and Table 4 (Example 37) communicated the four-factor priority system effectively by showing both the raw hazard ratings and the resulting normalized weights, allowing the reader to understand why certain rooms received priority over others. Figure 9's (Example 38) dual presentation bar charts compared four strategies and line graphs showing sweep time versus responder count, provided compelling visual evidence of the MILP model superiority over random, greedy, and manual-split approaches. Furthermore, all numerical charts had clear and readable axis labels. For Problem B, there were so many teams that presented excellent tables and figures with descriptive labels and explanations embedded in the accompanying section's text. Judges wanted to highlight the way Team #16517 used (Example 39) tables to summarize information. For example, Table 20 conveys the comparison of the important factors between hosting a Superbowl and the Olympics.

Creativity

Creativity is an important element of strong modeling and may appear at any stage of the process. This creativity may be reflected in how a problem is framed, how key factors are identified, or how models are constructed and adapted. It can also appear in the way teams combine methods, incorporate real-world considerations, or communicate their results. Judges value originality when it is purposeful. It is not about unnecessary complexity, but about choices that improve the model and its applications. The most successful teams demonstrate creativity by approaching the problem with curiosity and a willingness to explore ideas beyond the most obvious solution paths.

Emergency evacuation sweep problems like Problem A are challenging and involve reconciling competing objectives under uncertainty. Unlike optimization problems with clear single objectives, these problems require consideration of factors such as: minimizing total clear-

ance time, prioritizing vulnerable occupants, balancing responder workloads, and accounting for unpredictable hazards like spreading fire, structural collapse, and occupant behavior under stress. Creative modeling approaches move beyond deterministic models to capture the tail-risk (unlikely) scenarios that are important in safety-critical applications. Creativity comes from recognizing which real-world complexities need to be explicitly modeled (such as dynamic path real-time re-routing or fatigue affecting responder speed) versus which can be abstracted away; and from designing computational frameworks that remain tractable while preserving the stochastic, time-dependent nature of actual emergencies.

Paper 16603 (Example 40) demonstrates creativity through an innovative "Optimize-then-Simulate" hybrid framework that combines deterministic optimization with stochastic simulation. Rather than treating the evacuation sweep problem as purely deterministic or probabilistic, the team separates the problem into three computational stages: K-Means clustering for zone assignment, TSP (Traveling Salesperson Problem) heuristics for path planning, and Monte Carlo simulation with pathfinding for uncertainty modeling (referred to as A* modeling in the paper). They recognize that while optimal routes can be pre-computed under ideal conditions, actual emergency sweeps require dynamic adaptation to blockages, smoke, and rescue needs. The model's capability to distinguish between fire scenarios (higher baseline speed, smoke penalties, zero blockages) and earthquake scenarios (lower speed, structural blockages, clear visibility) through straightforward parameter toggling demonstrates an elegant design for reusability across emergency types.

For Problem B, the judges expected certain types of models (such as AHP, EWM, PCA and TOPSIS) to be used and were impressed when teams employed unique strategies or model types. Team 16785 (Example 41) used a two-stage machine learning framework to create their model. Team 16911 (Example 42) created three machine learning models to determine the best city to host in 2029. While Team 16521 (Example 43) used PCA and they added a geographic factor to their ecological footprint index to incorporate regional differences. Additionally, they inventively used a Borda count to establish a final ranking.

The judges also enjoyed seeing teams use novel techniques in other steps of the modeling process. For example, Team 16947 (Example 44) explained steps a city could take to improve their ability to be chosen as a sustainable host but then went further and picked a city that had ranked in the middle and showed how the city's ranking would improve if they applied some of the suggested strategies. Team 16739 (Example 45) employed ARIMA to predict the characteristics of the cities in 2029 to plug into their model.

Citations, References, AI Usage Report

Teams must give credit where credit is due. It is okay to use existing models found elsewhere, but teams that do this but do not cite, lose the trust of the judges. Teams that use existing models must cite their source(s) within the paper at the point they present the model and include a reference citation in the back of the paper. This is also true for all graphs and tables taken from literature. All data, figures, graphs, and tables that come from outside sources require documentation at the point in the paper that they appear. Lack of documentation will result in a lower designation. Teams should use "in line" documentation with footnotes or endnotes to give proper credit to outside sources.

With the increasing availability of AI tools, COMAP recognizes that teams may use these resources to support their modeling process and the preparation of their submissions. Any use of AI must be treated as an external source and clearly documented in an AI Use Report. This documentation should provide transparency into how the tool was used by including the excerpts of prompts and outputs. Teams have used AI tools in a variety of ways, such as gathering or summarizing data, exploring coding approaches, or generating potential modeling strategies. When used thoughtfully, these tools can support and enhance the modeling process. However, reliance on AI without careful evaluation can weaken a submission. Like any resource, AI outputs should be approached critically. Teams are responsible for any AI-generated content they choose to incorporate. Thoughtful and responsible use of AI strengthens both the credibility and quality of a submission.

Final Thoughts

On behalf of COMAP and the contest judges, we would like to again thank all advisors and students for their participation in the HiMCM. Each year the quality and level of mathematics demonstrated by the high school student teams amazes and impresses our judges. We truly enjoy reading all solution papers. Successful teams use a wide variety and level of mathematics. While teams using post-high school/undergraduate level mathematics are in a league of their own if they understand and can explain their work, teams using basic high school mathematics and much simpler approaches are often among our Outstanding designees. We encourage students of all levels to compete in future HiMCM competitions as well as our ICM/MCM contests targeted to undergraduates. To be successful, read the comments and guidance provided in this article, see the TIPS article on the COMAP website, and visit: www.comap.org/mathmodels to review previous problems.

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